

SKILL WEEK – 3

TASK 1

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
```

```
    char input[100];
```

```
    while (1) {
```

```
        printf("my_shell> ");
```

```
        if (fgets(input, sizeof(input), stdin) == NULL) {
```

```
            break;
```

```
        }
```

```
        input[strcspn(input, "\n")] = '\0';
```

```
        if (strcmp(input, "exit") == 0) {
```

```
            printf("Exiting shell...\n");
```

```
            break;
```

```
        }
```

```
        if (strlen(input) == 0) {
```

```
            continue;
```

```
        }
```

```
        printf("You entered: %s\n", input);
```

```
    }
```

```
    return 0;
}
```

```
nikith@NIKITH:~$ cd ~/OSSP
nikith@NIKITH:~/OSSP$ pwd
/home/nikith/OSSP
nikith@NIKITH:~/OSSP$ nano task1.c
nikith@NIKITH:~/OSSP$ gcc task1.c -o task1
nikith@NIKITH:~/OSSP$ ./task1
my_shell> ./task1
You entered: ./task1
my_shell> hello
You entered: hello
my_shell> my_shell> exit
Exiting shell...You entered: my_shell> exit
my_shell> |
```

TASK 2

```
#include <stdio.h>
```

```
#include <fcntl.h>
```

```
#include <unistd.h>
```

```
#include <stdlib.h>
```

```
int main() {
```

```
    int source, destination;
```

```
    char buffer[100];
```

```
    ssize_t bytesRead, bytesWritten;
```

```
    source = open("File1.txt", O_RDONLY);
```

```
    if (source < 0) {
```

```
        perror("Error opening File1.txt");
```

```

    return 1;
}

destination = open("File2.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

if (destination < 0) {
    perror("Error opening File2.txt");
    close(source);
    return 1;
}

while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0) {
    bytesWritten = write(destination, buffer, bytesRead);

    if (bytesWritten != bytesRead) {
        perror("Error writing to File2.txt");
        close(source);
        close(destination);
        return 1;
    }
}

if (bytesRead < 0) {
    perror("Error reading File1.txt");
}

close(source);
close(destination);

```

```
printf("File copied successfully!\n");
```

```
return 0;
```

```
}
```

```
nikith@NIKITH:~$ nano filecopy.c
nikith@NIKITH:~$ gcc filecopy.c -o filecopy
nikith@NIKITH:~$ ./filecopy
File copied successfully!
nikith@NIKITH:~$ cat File2.txt
Hello World!
This is File1.
This file is being copied using Linux system calls.
nikith@NIKITH:~$ diff File1.txt File2.txt
nikith@NIKITH:~$ |
```